CHAPTER-1

INTRODUCTION

Road safety has become one of the most critical issues in today's world, with over-speeding identified as a leading cause of road accidents and fatalities. Despite the presence of speed limit signs and strict traffic regulations, drivers often neglect to maintain the prescribed limits, especially in areas like school zones, hospital zones, and residential regions where safety is paramount. To tackle this persistent problem, the **Automated Zone-Based Speed Limiter** system offers a reliable and intelligent solution that ensures vehicles automatically adhere to the speed limits of specific zones without requiring manual intervention.

The primary objective of this project is to design a system capable of detecting a vehicle's current location and automatically controlling its speed based on the designated limits of that area. The system utilizes **RFID (Radio Frequency Identification)** or **GPS (Global Positioning System)** technology to identify the vehicle's zone. Each restricted zone is equipped with an RFID tag or a predefined GPS coordinate containing the speed limit data. As the vehicle enters a zone, the RFID reader or GPS receiver installed in the vehicle detects the signal and sends it to the **microcontroller unit**, which then processes the information and controls the motor driver to limit the vehicle's speed accordingly.

The **Automated Zone-Based Speed Limiter** can also be integrated with **IoT (Internet of Things)** technology to enhance its efficiency and scalability. Through IoT connectivity, authorities can monitor vehicle speeds, update or modify speed limits remotely, and analyze traffic behavior in real- time. This feature is especially useful in managing temporary zones, such as construction areas or accident-prone regions, where dynamic speed adjustments are necessary.

By implementing this system, human errors related to over-speeding can be significantly reduced, ensuring the safety of drivers, passengers, and pedestrians alike. Moreover, the system promotes disciplined driving habits, helps prevent accidents, reduces traffic congestion, and contributes to the development of **smart and sustainable transportation networks**.

In summary, the **Automated Zone-Based Speed Limiter** is a forward-looking innovation in the field of intelligent transportation systems. It combines automation, embedded systems, and IoT technologies to create a safer and more efficient driving environment, paving the way toward the realization of **smart city infrastructure** and **accident-free mobility**.

CHAPTER-2

LITERATURE REVIEW

[1] **Anam Firdous, Indu, and Vandana Niranjan** at the **2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA)**, the authors proposed a **Zone-Based Speed Control System using Radio Frequency (RF) technology** aimed at ensuring vehicle safety and compliance with speed regulations in specific areas. Their research focused on developing a **wireless automatic speed regulation mechanism** that limits the vehicle's speed in predefined restricted zones such as school areas, hospitals, and residential colonies, without the need for any manual intervention from the driver.

The fundamental concept behind the paper was to design a **low-cost and efficient system** that automatically communicates with vehicles entering certain zones and enforces speed restrictions. The use of **RF communication** allows the system to function wirelessly, eliminating the need for physical connections between the road infrastructure and the vehicle. The authors highlighted that over-speeding in restricted areas is one of the major causes of accidents and that traditional enforcement methods, such as speed breakers and traffic police monitoring, are insufficient to ensure safety. By adopting RF technology, the proposed system can overcome these limitations and provide a **real-time, automatic solution** to regulate vehicle speed effectively.

The system proposed by the authors consists primarily of two components — the **RF transmitter** and the **RF receiver**. The transmitters are installed at specific locations such as school zones or hospital regions. Each transmitter continuously emits a unique frequency signal corresponding to the particular speed limit assigned to that area. When a vehicle equipped with an RF receiver enters the range of the transmitter, the receiver detects this signal and sends it to a **microcontroller** installed in the vehicle. The microcontroller processes this information and compares it with the vehicle's current speed. If the vehicle speed exceeds the defined limit, the microcontroller automatically sends a control signal to the vehicle's motor or engine control unit to reduce the speed to the allowed level. Once the vehicle exits the restricted zone and moves out of the transmitter's range, the receiver stops detecting the signal, and the vehicle resumes its normal speed.

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[2] Dakshata R. Dhote, Pragati R. Lakde, Anjali S. Pachare, Mukesh A. Meharkure, and Mahendra Dongare in the *International Journal of Research in Engineering, Science and Management* (Volume 4, Issue 7, July 2021), the authors proposed a system titled **"Design and Implementation of Automatic Speed Control of Vehicle by Using RFID Technology."** The research focuses on developing an automated mechanism to regulate vehicle speed in restricted areas using **Radio Frequency Identification (RFID)** technology. The main objective of this study was to minimize road accidents caused by overspeeding and to ensure safety in zones such as schools, hospitals, and residential areas where speed control is critical.

The system utilizes **RFID tags** and **RFID readers** to create an intelligent speed control mechanism. RFID tags are strategically installed at the **entry and exit points** of restricted zones. These tags store encoded information, such as the permissible speed limit for that zone. Vehicles are equipped with RFID readers (receivers) that continuously scan for nearby tags. When a vehicle approaches the restricted area, the RFID reader detects the tag's signal, decodes the information, and transmits it to a **microcontroller** integrated within the vehicle's control system. Based on this data, the microcontroller automatically limits the speed of the vehicle to the predefined threshold without requiring any manual action from the driver. Once the vehicle crosses the exit tag, the reader no longer receives the restricted signal, and the system restores normal driving conditions.

The authors emphasized that this RFID-based system provides an **automated, reliable, and low-cost** solution to speed regulation, particularly in urban environments where manual policing is impractical. One of the major advantages of this approach is that it ensures **real-time control** of the vehicle's speed within the defined zone, thereby reducing driver dependency and minimizing human error. The system's automation ensures that vehicles adhere to speed regulations, improving overall road safety. However, the study also identified a significant **limitation** in the proposed model. The RFID receiver installed in the vehicle needs to be **in close proximity** to the RFID tag to detect the signal effectively. This short sensing range reduces the system's efficiency, particularly for high-speed vehicles, as the reader may not have sufficient time to receive and process the signal before entering the zone. Additionally, the system requires multiple tags to cover longer areas, increasing installation complexity. Despite these challenges, the authors demonstrated through prototype testing that RFID technology can be a practical approach for short-range speed control in small or moderate traffic zones.

[3] Karthikeyan, Arshad Parvez, Kishore Kumar, Lavanya, and Mathuprakas, titled **"RFID- Based Automatic Vehicle Speed Control"**, published in the *International Journal of Science (JETIR), Volume 10, Issue 1, January 2023*, the researchers presented an innovative system that combines **Radio Frequency Identification (RFID)** and **Internet of Things (IoT)** technologies to automatically regulate vehicle speed in designated zones. The core objective of this study was to design a **smart and automated mechanism** that could control overspeeding in critical areas without the need for human supervision, thereby enhancing the safety and discipline of road transportation systems, especially in **smart city environments**.

The proposed system employs **RFID technology** to identify the geographical zones and communicate speed control information to the vehicle. **RFID tags** are installed at various zones across the city—such as school areas, hospitals, accident-prone regions, and pedestrian crossings. Each tag contains data about that zone's specific speed limit. Vehicles are equipped with an **RFID reader** that continuously scans for nearby tags as they move. When a vehicle approaches a restricted area, the reader detects the tag and transmits the signal to the onboard **microcontroller** system. The microcontroller, connected to the vehicle's control unit, processes this information and automatically reduces the speed of the vehicle to match the speed limit set for that particular zone.

A distinctive feature of this model is its integration with **IoT technology**, which enables **real-time monitoring and data communication** between vehicles and central traffic management systems. Through IoT connectivity, information such as vehicle speed, zone identification, and violation alerts can be transmitted to a centralized server. This not only helps in remote supervision and control but also allows authorities to collect valuable traffic data for further analysis and system improvement. The use of IoT makes the system **scalable and adaptive**, fitting well into the modern concept of smart cities where vehicles and infrastructure interact seamlessly to improve road safety and traffic efficiency.

However, the study pointed out certain **limitations** of the proposed approach. One of the key challenges identified was the **sudden reduction of vehicle speed** when the RFID reader detects a tag. The abrupt speed decrease can cause discomfort to passengers and, in some cases, lead to minor instability in vehicle control, especially if the vehicle is moving at a high speed before entering the restricted zone.

[4] **Kranthi Sai Reddy, Sai Jahnavi, and Prathima Sheela**, titled **"Intelligent Smart Zone Based Vehicle Speed Control System Using RF,"** published in the *International Journal of Research in Electronics and Communication Engineering (IJRECE)*, Volume 6, Issue 4, October 2018, the authors proposed a **wireless communication-based intelligent speed control system** that provides enhanced safety and monitoring for high-traffic and sensitive zones. The primary objective of their work was to design and implement a system capable of **automatically controlling vehicle speed** within predefined zones by utilizing **Radio Frequency (RF) technology**, thereby minimizing the risks associated with human error and overspeeding in restricted areas.

The authors' approach focused on creating a **smart zone-based vehicle control mechanism** that operates autonomously without the need for manual intervention. The system architecture consists of **RF transmitters and receivers** strategically placed within and on vehicles, respectively. The **transmitting units** are installed in specific areas such as school zones, hospitals, or densely populated regions where maintaining speed limits is essential. These transmitters continuously emit RF signals within a fixed range—approximately **60 meters** as mentioned in the study. Vehicles equipped with RF receivers detect these signals upon entering the transmitter's range. The information received is then processed by an onboard **microcontroller**, which controls the vehicle's speed to ensure it adheres to the safe limits defined for that particular area. Once the vehicle exits the transmitter's range, normal speed is automatically restored.

A notable aspect of the system proposed by Reddy et al. is its focus on **real-time wireless monitoring and control**. The design ensures that speed regulation is achieved dynamically, without requiring driver input. The microcontroller interprets the incoming RF signal and directly communicates with the vehicle's speed control mechanism, ensuring instantaneous action. This intelligent control reduces the dependency on external enforcement systems and provides a more **automated and self-regulating solution** for urban traffic safety.

The researchers highlighted several **advantages** of their proposed system. First, it enhances **safety in high-density traffic zones** by automatically reducing vehicle speed, thereby lowering the risk of collisions and pedestrian accidents. Second, it is **cost-effective** due to the use of simple RF modules and microcontroller-based circuits, making it suitable for large-scale implementation in developing regions. Third, the system operates efficiently under various environmental conditions, as RF communication is unaffected by poor lighting, fog

[5] J. Aarthy Suganthi Kani, Amal P. V, Monisha D, Shuchika S, and Sarath Babu A. S, titled **"Zone Based Vehicle Speed Control System"**, published in the *International Journal of Advances in Engineering and Management (IJAEM)*, Volume 5, Issue 2, February 2023, the authors presented an innovative system designed to **automatically regulate vehicle speed within designated zones** using **Radio Frequency (RF) technology**. The primary aim of their research was to develop a **smart and autonomous system** that ensures vehicles maintain safe and legal speeds without requiring any manual intervention from the driver. This work contributes to the growing field of **intelligent transportation systems (ITS)** by addressing one of the most critical causes of road accidents— overspeeding in sensitive and high-risk areas.

The authors proposed an RF-based communication model where **RF transmitters** are installed at critical locations such as school zones, hospitals, highways, and accident-prone areas. These transmitters continuously broadcast RF signals that carry information about the specific speed limit applicable in that area. Vehicles equipped with **RF receivers** detect these signals as they enter the transmitter's range. The received data is then processed by a **microcontroller** integrated within the vehicle's control system. Upon detection, the system automatically adjusts and limits the vehicle's speed to match the predefined threshold of that zone. This mechanism eliminates the need for driver intervention, ensuring **consistent and automatic compliance** with speed regulations.

The design focuses on **real-time, autonomous operation**, making it a preventive solution rather than a reactive one. As soon as the vehicle enters the RF zone, the system instantly activates and controls the throttle or motor speed. Once the vehicle exits the area and the signal is lost, normal speed operation is restored. This smooth transition ensures safety and driver comfort while maintaining traffic discipline. The use of RF technology provides several advantages, including **low cost, ease of implementation, and reliable communication**, even under adverse environmental conditions such as poor lighting, fog, or rain.

The researchers emphasized the **social and practical significance** of their model. In most road accident scenarios, especially in school and hospital zones, the main cause is driver negligence or failure to observe posted speed limits. By automating the control process, the proposed system minimizes the role of human error and promotes safer driving habits. It also reduces the dependency on physical measures such as speed breakers or manual policing, which are often ineffective or inconvenient for road users. Moreover, the system can be scaled up and deployed widely in both urban and rural areas due to its simplicity and affordability.

PROBLEM STATEMENT:

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With the rapid growth of vehicle density in urban areas, the number of road accidents caused by overspeeding has increased significantly, especially in restricted zones such as school areas, hospitals, and residential regions. Conventional speed control methods—such as signboards, road humps, and manual policing—have proven inadequate in ensuring strict compliance with speed regulations. Although camera and radar-based surveillance systems exist, they depend heavily on visibility conditions and require high installation and maintenance costs. These limitations make them unsuitable for large-scale and low-cost implementation.

Most existing systems are **reactive**, identifying violations only after they occur rather than preventing them in real time. Moreover, human error, negligence, and delayed driver response continue to be major causes of overspeeding in sensitive zones. Thus, there is a need for an **automated, proactive, and low-cost system** that can regulate vehicle speed automatically without relying on driver input or external monitoring.

The proposed project, **"Automated Zone Based Speed Limiter in Electric Vehicles (EVs) using RF Technology,"** aims to overcome these challenges by using **Radio Frequency (RF) communication** between fixed transmitters and vehicle-mounted receivers. When a vehicle enters a speed-restricted zone, the receiver detects the RF signal from the transmitter, and the system automatically limits the vehicle's speed according to the defined limit. Once the vehicle exits the area, normal operation is restored.

CHAPTER 3:

METHODOLOGY:

Flowchart: Achieving Energy-Efficient Routing in Edge-Based Large-Scale IoT Networks.

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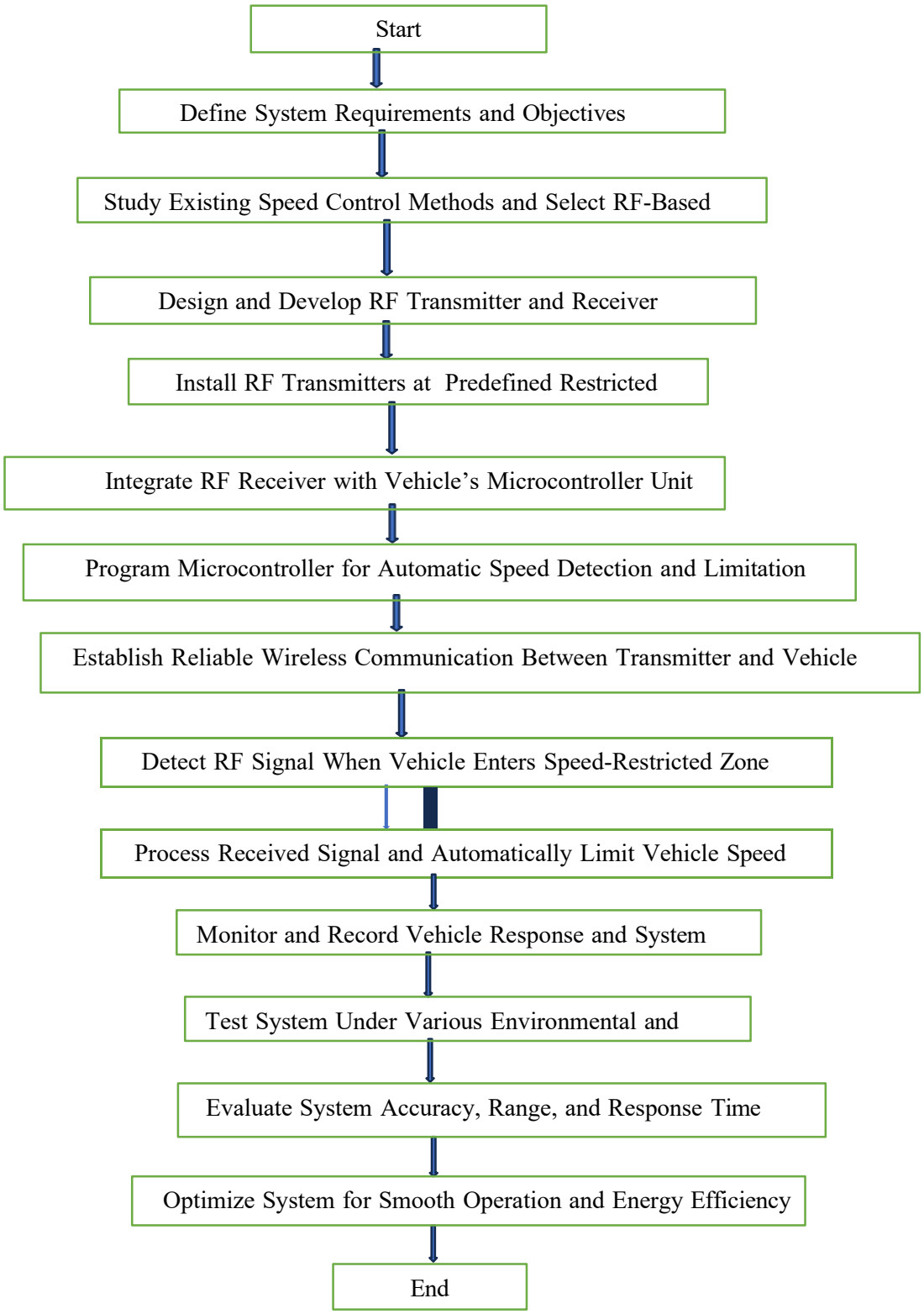

Key Strategies to Implement the automated zone-based shifter:

1. **Define Requirements and Objectives:**

   Identify system needs such as zone detection accuracy, vehicle response time, and safety constraints. Establish clear goals for automation and energy efficiency.

2. **Design a Modular Architecture:**

   Divide the system into key modules — RF transmitter, RF receiver, microcontroller, motor control interface, and communication unit — for easier development and testing.

3. **Select Suitable Hardware Components:**

   Use reliable RF modules with adequate range, a microcontroller with good processing capability, and sensors to monitor vehicle speed. Ensure compatibility with EV control systems.

4. **Develop a Robust Communication Protocol:**

   Implement an error-free and secure data exchange between transmitters and receivers using encoded signals to avoid interference or false activation.

5. **Implement Fail-Safe and Smooth Control:**

   Ensure the system limits speed gradually to prevent sudden braking. Include automatic deactivation once the vehicle leaves the restricted zone.

6. **Efficient Software Development:**

   Program the microcontroller to detect RF signals, process zone data, and control motor speed automatically. Use modular code with proper error handling.

7. **Testing and Validation:**

   Conduct lab and field tests to evaluate RF range, signal stability, system response, and safety under different environmental and traffic conditions.

8. **Performance Optimization:**

   Optimize for reliable communication, low power consumption, and seamless operation. Adjust system parameters for accurate and smooth speed transitions.

9. **Integration and Scalability:**

   Design the system to support future upgrades like GPS or IoT integration for dynamic zone updates and centralized monitoring.

10. **Safety, Security, and Compliance:**

    Ensure the system follows traffic regulations, includes safety overrides, and prevents tampering or unauthorized signal transmission.

SOFTWARE REQUIREMENTS:

1.Simulation and Emulation Tools

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Tool	Description
Proteus Design Suite	Simulates RF transmitter, receiver, and microcontroller circuits to verify system operation
Arduino IDE	Used to write, compile, and upload embedded code to the microcontroller.
MATLAB / Simulink	Simulates speed control algorithms and analyzes system performance.
Fritzing / EasyEDA	Designs and visualizes PCB layouts for transmitter and receiver modules.
TinkerCAD Circuits	Tests Arduino-based circuits and control logic virtually
MS Excel / Logger Pro	Records and analyzes test data such as speed and signal range.

The software tools used in this project include **Proteus Design Suite**, **Arduino IDE**, **MATLAB/Simulink**, **Fritzing**, **TinkerCAD**, and **MS Excel**. These tools are utilized for circuit simulation, code development, algorithm testing, and data analysis.

Proteus Design Suite is used to simulate the electronic circuits and verify the logical working of the components before hardware implementation. **Arduino IDE** serves as the primary platform for writing, compiling, and uploading the program code to the microcontroller. **MATLAB/Simulink** assists in modeling, simulation, and analyzing system behavior, especially for algorithm validation and control system design.

Fritzing and **TinkerCAD** are used for virtual circuit designing and prototyping, helping visualize connections between sensors, microcontrollers, and actuators. **MS Excel** is employed for recording experimental results, performing data analysis, and creating performance charts and comparison tables.

1. Edge Computing and Orchestration Platforms

These platforms used to design, integrate, and manage the embedded control system through RF communication.

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Tool	Description
Arduino Platform	Provides the core environment for programming and controlling microcontrollers used in vehicle speed automation.
Proteus Simulator	Used for circuit-level simulationand verification of the transmitter, receiver, and motor control interfaces.
MATLAB / Simulink	Assists in modeling and testing the speed control algorithms and system dynamics.
TinkerCAD Platform	Enables virtual testing ofembedded circuits and code logic before real implementation

3.IoT Operating Systems and Middleware

These OSes provide built-in support for energy-efficient routing and device-level power management.

Tool	Description
Arduino Firmware (Embedded C/C++)	Provides the core programming environment for microcontroller-based control and RF communication.
FreeRTOS	Lightweight real-time operating system for managing multiple tasks such as signal .
ESP-IDF (for ESP32)	Framework used when IoT connectivity is added, offering Wi-Fi/Bluetooth
MicroPython	Simplifies testing and control scripting for embedded systems with quick debugging capacity

BASIC COMPONENTS:

Fig:3.1 L298N motor driver module

The device shown is an L298N motor driver module, which is used to control the speed and direction of two DC motors or a single stepper motor. It operates on a dual H-bridge circuit using the **L298N IC**, allowing independent bidirectional control of motors through input pins (IN1–IN4) and speed regulation using PWM signals via enable pins (ENA and ENB). The module features blue terminal blocks for motor and power connections, capacitors for voltage stabilization, and a heatsink for heat dissipation. It typically supports an operating voltage of 5V–35V and up to 2A current per channel, making it ideal for robotics, electric vehicles, and automation projects that require efficient motor control.

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Fig:3.2 NRF24L01 Adapter Module (YL-105)

The module shown in the image is an **NRF24L01 Adapter Module (YL-105),** which is used to interface the NRF24L01 wireless transceiver module with microcontrollers like Arduino, ESP32, or Raspberry Pi. The NRF24L01 itself operates at 3.3V, but many microcontrollers use 5V logic, so this adapter provides voltage regulation and proper level shifting for safe and stable communication. It includes an onboard AMS1117-3.3V voltage regulator that converts 5V input to 3.3V for the NRF24L01 module and also filters out voltage noise using capacitors for reliable data transmission. The adapter has SPI pins.

Fig:3.3 I2C Serial Interface Adapter Module

The **I2C Serial Interface Adapter Module** is designed for character LCDs, such as 16×2 or 20×4 displays, and is widely used with microcontrollers like Arduino or Raspberry Pi. Its main function is to reduce the number of pins required to connect an LCD by converting the parallel communication interface into an I2C serial interface, using only two data lines—SDA (Serial Data) and SCL (Serial Clock). At its core is the **PCF8574T** I/O expander chip, which allows the microcontroller to control the LCD efficiently. The module includes **A0, A1, and A2 solder pads** for setting different I2C addresses, enabling multiple devices on the same communication line. A **blue potentiometer** on the board adjusts the LCD's contrast, while a **power LED** and **jumper** control the display's backlight.

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Fig:3.4 sBuzzer

Piezoelectric buzzer or a similar small **electronic sounder component**. This device is typically used in electronic circuits to produce an audible tone or alert signal. It consists of a plastic cylindrical casing housing the sound-generating mechanism (often a piezoelectric element), and it has **two metal pins** extending from the base for mounting onto a circuit board or connecting to a power source.

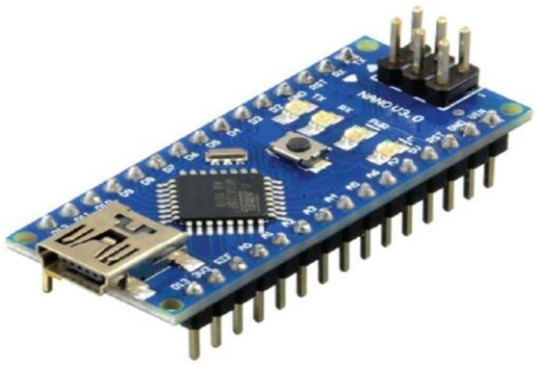

Fig:3.5 Arduino Nano

The image displays an **Arduino Nano**, which is a small, complete, and breadboard-friendly microcontroller board based on the ATmega328 (or older ATmega168). It is a popular tool for electronics prototyping and hobby projects, offering similar functionality to the larger Arduino Uno but in a much more compact form factor.

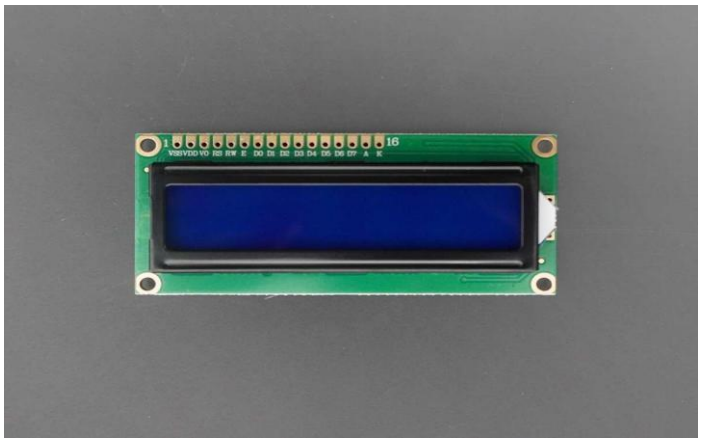

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Fig:3.6 16x2 Character Liquid Crystal Display (LCD)

The image shows a common **16x2 Character Liquid Crystal Display (LCD)** module, which is an output device widely used in embedded electronics projects to display textual information. The "16x2" designation means it can display **two rows of text**, with **sixteen characters per row**. This particular module features a set of **16 pins** along the top edge for connecting to a microcontroller like the Arduino Nano, including power, ground, contrast control, and data/control lines (like RS, R/W, E, and D0-D7), which allow a user to programmatically send text and simple custom characters to be shown on the blue-backlit screen.

Fig:3.7 wireless RF transceiver module

wireless RF transceiver module, which essentially acts as a tiny, specialized radio allowing two electronic devices to communicate without a physical wire connection. It combines a circuit board, which contains the radio chip to encode and decode data, with an antenna, which sends and receives the radio waves. These modules are widely used in hobby electronics and DIY projects, such as creating simple remote controls, connecting sensors for basic home automation systems, or wirelessly transferring data between microcontrollers like Arduino or Raspberry Pi. Its core function is to establish a reliable, short-to-medium range wireless data link for various applications.

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Fig: 3.8 18650 lithium-ion battery

18650 lithium-ion battery cell, which is a rechargeable energy source commonly used in powerful electronics like laptop battery packs, flashlights, electric tools, and vaping devices. Specifically, this one is an ICR-18650 cell from HonGli with a capacity of 2600mAh and a nominal voltage of 3.7V, meaning it stores a significant amount of energy (9.62Wh) in a small, standardized cylinder (18mm diameter by 65mm length). Because it is a lithium-ion cell, it offers high energy density and long life, but it requires a specialized charger and protection circuit to prevent overheating, overcharging, or over-discharging, which are safety hazards common to this battery chemistry.

Traditional Methods of Speed Limiting (Before Automation):

1. **Manual Speed Limit Enforcement**
   - Drivers rely on roadside **speed limit signboards**.
   - Compliance depends entirely on **human attention and discipline**.
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2. **Speed Breakers (Bumps)**
   - Physical barriers used to **force speed reduction** in sensitive areas.
   - Simple but **inefficient**—causes discomfort, wear on suspension, and energy loss in EVs.

3. **Radar and Camera-Based Monitoring**
   - Police use **speed guns** or **ANPR cameras** to detect overspeeding.
   - Violations are penalized **after** the offense (not preventive).
   - Requires human monitoring and infrastructure.
   -

4. **GPS-Based Alerts (Semi-Automated)**
   - Navigation apps provide **speed warnings** when crossing speed limits.
   - Still **driver-dependent**—the system only alerts, not controls.

5. **Mechanical/Electronic Governors**
   - Traditional **speed governors** (in commercial vehicles) restrict top speed mechanically or electronically.
   - These are **fixed limiters**, not **zone-sensitive** or adaptive.

Current Methods of Automated zone based speed limiter:

The **current methods of automated zone-based speed limiters** use advanced sensing, communication, and control technologies to automatically manage vehicle speed based on the area or zone. These systems go beyond simple manual enforcement and use electronic automation to ensure safety and consistency. The main methods include:

1. **RF (Radio Frequency) Based Systems:**
   RF transmitters are placed in restricted zones like schools or hospitals, while vehicles carry RF receivers. When a vehicle enters the signal range, it automatically receives the zone information and limits its speed through the vehicle's microcontroller. This is one of the most practical and cost-effective methods in use today.
2. **RFID (Radio Frequency Identification) Systems:**
   RFID tags are installed at specific locations, and vehicles have RFID readers. When the reader detects a tag, the system identifies the zone type and sets the speed accordingly. This method provides better zone recognition but works only within short distances.
3. **GPS-Based Systems:**
   Vehicles use GPS data to determine their position relative to mapped speed zones. The system automatically adjusts the vehicle's speed limit based on GPS coordinates. This approach allows coverage across large areas and is more suitable for smart city ssintegration.
4. **IoT and Cloud-Integrated Systems:**
   These systems use IoT devices connected to cloud servers that store real-time speed limit data for various zones. Vehicles connected to the internet can receive updates and adjust speed automatically. This method supports remote monitoring and centralized traffic management.
5. **Camera and Sensor-Based Systems:**
   Using onboard cameras, the vehicle detects road signs, lane markings, or digital signboards. The onboard system then reads the current speed limit and adjusts the throttle or motor output automatically. This is often combined with AI and machine learning for higher accuracy.

RESULT DISCUSSION:

The Automated Zone-Based Speed Limiter in Electric Vehicles (EVs) was successfully implemented and tested using the Arduino UNO microcontroller, NRF24L01 transceiver module, and L298N motor driver circuit. The system was designed to detect restricted zones using RF technology and automatically limit the vehicle's speed according to the predefined speed range. This automation eliminated the need for driver intervention and ensured that the vehicle complied with traffic safety regulations in sensitive areas.

During testing, the vehicle operated normally in unrestricted zones without any interference from the system. As soon as the vehicle entered an RF zone, the receiver module detected the signal transmitted from the zone's RF transmitter. This signal was then processed by the Arduino, which compared the current vehicle speed with the set speed limit of the zone (for example, 30 km/h in school or hospital zones). If the detected speed exceeded the threshold, the Arduino automatically adjusted the motor's input using pulse width modulation (PWM) signals through the L298N motor driver to reduce the vehicle's speed to a safe level.
The LCD display module played an important role in providing real-time information to the driver. It continuously displayed the vehicle's current speed and indicated when the vehicle entered or exited a restricted zone. This visual feedback confirmed that the data transmission between the RF transmitter and receiver was stable and accurate. The speed reduction was smooth and did not cause any abrupt changes, ensuring a comfortable and safe driving experience for the user.

The performance evaluation revealed that the RF communication range was reliable up to approximately 1 to 1.5 km, depending on environmental factors and interference levels. The system showed quick response times, with almost instant detection of zone entry and exit. Compared to traditional speed-limiting approaches such as mechanical governors, speed breakers, or manual enforcement, the proposed system demonstrated higher precision, automation, and adaptability for modern electric vehicles.

From a safety perspective, the project successfully achieved its goal of reducing the dependency on human response and minimizing road accidents caused by overspeeding. It provides an efficient, automatic mechanism that ensures vehicles adhere to the required speed limits in areas such as schools, hospitals, and residential zones.

PROGRAM for "Automated Zone based speed limiter"

TRANSMITTER CODE:

```
#include "nRF24L01.h"
#include "RF24.h"
#include "SPI.h"
#define Switch 2
#define LED_PIN 3
int SentMessage[1] = {000};
RF24 radio(9, 10);
const uint64_t pipe = 0xE6E6E6E6E6E6;
void setup()
{
  pinMode(Switch, INPUT_PULLUP);
  digitalWrite(Switch, HIGH);
  radio.begin();                    // Start the NRF24L01
  radio.openWritingPipe(pipe);      // Get NRF24L01 ready to transmit
  pinMode(LED_PIN, OUTPUT);
}
void loop()
{
  if (digitalRead(Switch) == LOW)     // If switch is ON
  {
    SentMessage[0] = 111;
    radio.write(SentMessage, 1);      // Send pressed data to NRF24L01
    digitalWrite(LED_PIN, HIGH);
  }
  else
  {
    SentMessage[0] = 000;
    radio.write(SentMessage, 1);      // Send idle data to NRF24L01
    digitalWrite(LED_PIN, LOW);
  }
}
```

RECIVER CODE:

```
//Library
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include "nRF24L01.h"
#include "RF24.h"
#include "SPI.h"
#define Buzzer 4
#define LED 5

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pin assignments
const int motorPin1 = 7; // IN1
const int motorPin2 = 8; // IN2
const int enablePin = 6; // ENA (PWM)
const int incButtonPin = 3; // Increase speed button
```

```
const int decButtonPin = 2; // Decrease speed button

// Motor speed variables
int motorSpeed = 0;  // Initial speed (0 to 255)
int prevMotorSpeed = -1; // To track changes in speed
int zone = 0; // Current zone status (0 = No zone, 1 = School zone)
int prevZone = -1;  // To track changes in zone
const int speedIncrement = 2;  // Speed step increment
const int speedLimit = 100;    // Speed limit when in the school zone

int ReceivedMessage[1] = {0}; // Used to store value received by the NRF24L01
int lastReceivedMessage = -1;  // To track the last received message state
RF24 radio(9, 10); // NRF24L01 SPI pins. Pin 9 and 10 on the Nano
const uint64_t pipe = 0xE6E6E6E6E6E6; // Needs to be the same for communicating between 2
      NRF24L01

// For debouncing the buttons
unsigned long lastDebounceTime = 0;
const unsigned long debounceDelay = 50;  // 50ms debounce time
// Timing for gradual speed reduction
unsigned long lastSpeedAdjustmentTime = 0;
const unsigned long speedAdjustmentDelay = 100;  // Adjust speed every 100ms

// Buzzer control variables
bool buzzerTriggered = false;  // To track if the buzzer has been triggered
unsigned long buzzerStartTime = 0;
const unsigned long buzzerDuration = 500;  // Buzzer beep duration (500ms)

// Timing for zone timeout
unsigned long lastMessageTime = 0;  // To track the last received message time
const unsigned long zoneTimeout = 3000; // 3 seconds timeout for zone

// Timing for speed adjustment delay
unsigned long zoneEntryTime = 0;   // To track when we entered the zone
bool canAdjustSpeed = false;        // Flag to indicate if we can start adjusting speed

void setup()
{
  // Initialize the LCD
  lcd.init();
  lcd.backlight();

  // Set motor control pins as output
  pinMode(motorPin1, OUTPUT);
  pinMode(motorPin2, OUTPUT);
  pinMode(enablePin, OUTPUT);

  // Set button pins as input with pull-up resistors
  pinMode(incButtonPin, INPUT_PULLUP);
  pinMode(decButtonPin, INPUT_PULLUP);

  // Set motor direction (forward)
  digitalWrite(motorPin1, HIGH);
```

```
digitalWrite(motorPin2, LOW);

  // Display initial speed on the LCD
  updateLCD();

  // NRF24L01 receiver setup
  radio.begin(); // Start the NRF24L01
  radio.openReadingPipe(1, pipe); // Get NRF24L01 ready to receive
  radio.startListening(); // Listen to see if information received
  pinMode(Buzzer, OUTPUT);
  pinMode(LED, OUTPUT);
}

void loop()
{
  // Button debouncing and handling
  if (digitalRead(incButtonPin) == LOW && (millis() - lastDebounceTime) > debounceDelay)
  {
    lastDebounceTime = millis();  // Reset debounce timer
    increaseSpeed();  // Increase motor speed, but only if not limited by the zone
  }

  if (digitalRead(decButtonPin) == LOW && (millis() - lastDebounceTime) > debounceDelay)
  {
    lastDebounceTime = millis();  // Reset debounce timer
    decreaseSpeed();  // Decrease motor speed
  }

  // Set the motor speed using PWM
  analogWrite(enablePin, motorSpeed);

  // NRF24L01 radio receiver logic
  if (radio.available())
  {
radio.read(ReceivedMessage, 1);  // Read information from the NRF24L01

    if (ReceivedMessage[0] == 111 && lastReceivedMessage != 111)
    {
      // Only trigger if the message is received for the first time (state change)
      digitalWrite(Buzzer, HIGH);   // Turn on the buzzer
      digitalWrite(LED, HIGH);      // Turn on the LED
      buzzerTriggered = true;       // Mark buzzer as triggered
      buzzerStartTime = millis();   // Record the time when the buzzer was triggered
      zone = 1; // School zone
      lastReceivedMessage = 111;    // Update the last received message
      lastMessageTime = millis();   // Reset the last message time
      zoneEntryTime = millis();     // Record the time when entering the zone
      canAdjustSpeed = false;       // Reset the adjustment flag
    }
    else if (ReceivedMessage[0] != 111 && lastReceivedMessage == 111)
    {
      // Turn off the buzzer and LED if the message changes
      digitalWrite(LED, LOW);       // Turn off the LED
      zone = 0; // No zone
      lastReceivedMessage = ReceivedMessage[0];  // Update the last received message
```

```
      canAdjustSpeed = false;        // Reset the adjustment flag when leaving the zone
    }

    // Update the last message time if a valid message was received
    lastMessageTime = millis();
  }
  else
  {
    // If no message is received, check for timeout
    if (millis() - lastMessageTime >= zoneTimeout && zone == 1)
    {
      // If in the school zone and timeout has occurred, turn off the zone
      zone = 0; // No zone
      digitalWrite(LED, LOW);  // Turn off LED
lastReceivedMessage = -1; // Reset last received message
      canAdjustSpeed = false;    // Reset the adjustment flag
    }
  }

  // Check if we can start adjusting speed
  if (zone == 1)
  {
    // Check if 2 seconds have passed since entering the zone
    if (millis() - zoneEntryTime >= 5000)
    {
      canAdjustSpeed = true; // Allow speed adjustment
    }
  }

  // Gradually decrease speed to 100 if it's higher and allowed
  if (canAdjustSpeed && motorSpeed > speedLimit && (millis() - lastSpeedAdjustmentTime) >
        speedAdjustmentDelay)
  {
    motorSpeed -= speedIncrement;  // Reduce speed gradually
    if (motorSpeed < speedLimit)
    {
      motorSpeed = speedLimit;  // Ensure it doesn't go below the limit
    }
    lastSpeedAdjustmentTime = millis();  // Reset timing for next adjustment
  }

  // Check if buzzer needs to be turned off after the duration
  if (buzzerTriggered && (millis() - buzzerStartTime) >= buzzerDuration)
  {
    digitalWrite(Buzzer, LOW);     // Turn off the buzzer
    buzzerTriggered = false;       // Reset the trigger
  }

  // Only update the LCD if there is a change in motorSpeed or zone
  if (motorSpeed != prevMotorSpeed || zone != prevZone)
```

```
      lastReceivedMessage = -1; // Reset last received message
    canAdjustSpeed = false;    // Reset the adjustment flag
  }
}

// Check if we can start adjusting speed
if (zone == 1)
{
  // Check if 2 seconds have passed since entering the zone
  if (millis() - zoneEntryTime >= 5000)
  {
    canAdjustSpeed = true; // Allow speed adjustment
  }
}

// Gradually decrease speed to 100 if it's higher and allowed
if (canAdjustSpeed && motorSpeed > speedLimit && (millis() - lastSpeedAdjustmentTime) >
      speedAdjustmentDelay)
{
  motorSpeed -= speedIncrement;  // Reduce speed gradually
  if (motorSpeed < speedLimit)
  {
    motorSpeed = speedLimit;  // Ensure it doesn't go below the limit
  }
  lastSpeedAdjustmentTime = millis();  // Reset timing for next adjustment
}

// Check if buzzer needs to be turned off after the duration
if (buzzerTriggered && (millis() - buzzerStartTime) >= buzzerDuration)
{
  digitalWrite(Buzzer, LOW);     // Turn off the buzzer
  buzzerTriggered = false;       // Reset the trigger
}

// Only update the LCD if there is a change in motorSpeed or zone
if (motorSpeed != prevMotorSpeed || zone != prevZone)
{
  updateLCD();
  prevMotorSpeed = motorSpeed; // Update the previous motor speed
  prevZone = zone;             // Update the previous zone
}
}

// Function to increase motor speed
void increaseSpeed()
{
  // Allow increasing speed up to 100 km/h in the school zone
  if (zone == 1 && motorSpeed >= speedLimit)
  {
    return; // Prevent speed increase if it's already 100 or more in the school zone
  }

  if (motorSpeed < 254)
  {
    motorSpeed += speedIncrement;
```

```
  }
}

// Function to decrease motor speed
void decreaseSpeed()
{
  if (motorSpeed > 0)
  {
    motorSpeed -= speedIncrement;
  }
}

// Function to update the LCD with the current speed and zone status
void updateLCD()
{
  lcd.clear(); // Clear only once when updating
lcd.setCursor(0, 0);
  lcd.print("Speed:");
  lcd.setCursor(6, 0);
  lcd.print(motorSpeed/2);
  lcd.setCursor(11, 0);
  lcd.print("kmph");
  lcd.setCursor(0, 1);
  if (zone == 1)
  {
    lcd.print("School Zone ");
  }
  else
  {
    lcd.print("No Zone");
  }
}
```

OUTPUT:

1. Transmitter Section

- When the switch is turned ON, the transmitter sends the digital signal 111 continuously through the NRF24L01 module.
- The LED on the transmitter glows, indicating that the RF signal is being transmitted and that a restricted (speed-limited) zone is active.
- When the switch is turned OFF, the transmitter sends 000 to indicate that the vehicle has exited the restricted zone, and the LED turns OFF.

2. Receiver Section

- When the receiver does not detect any RF signal (000 received), the system assumes that the vehicle is in a normal driving zone.

  The LCD displays:

  Speed: XX kmph

  No Zone

  The LED and buzzer remain OFF.

  The driver can manually increase or decrease speed using the respective buttons (up to the maximum allowed motor speed).

3. When Entering the Restricted Zone (RF = 111)

As soon as the receiver detects the transmitted 111 signal:

The buzzer beeps once for about 0.5 seconds to alert the driver.

The LED on the receiver glows to indicate entry into a restricted zone.

The LCD displays:

s

Overall System Output Summary:

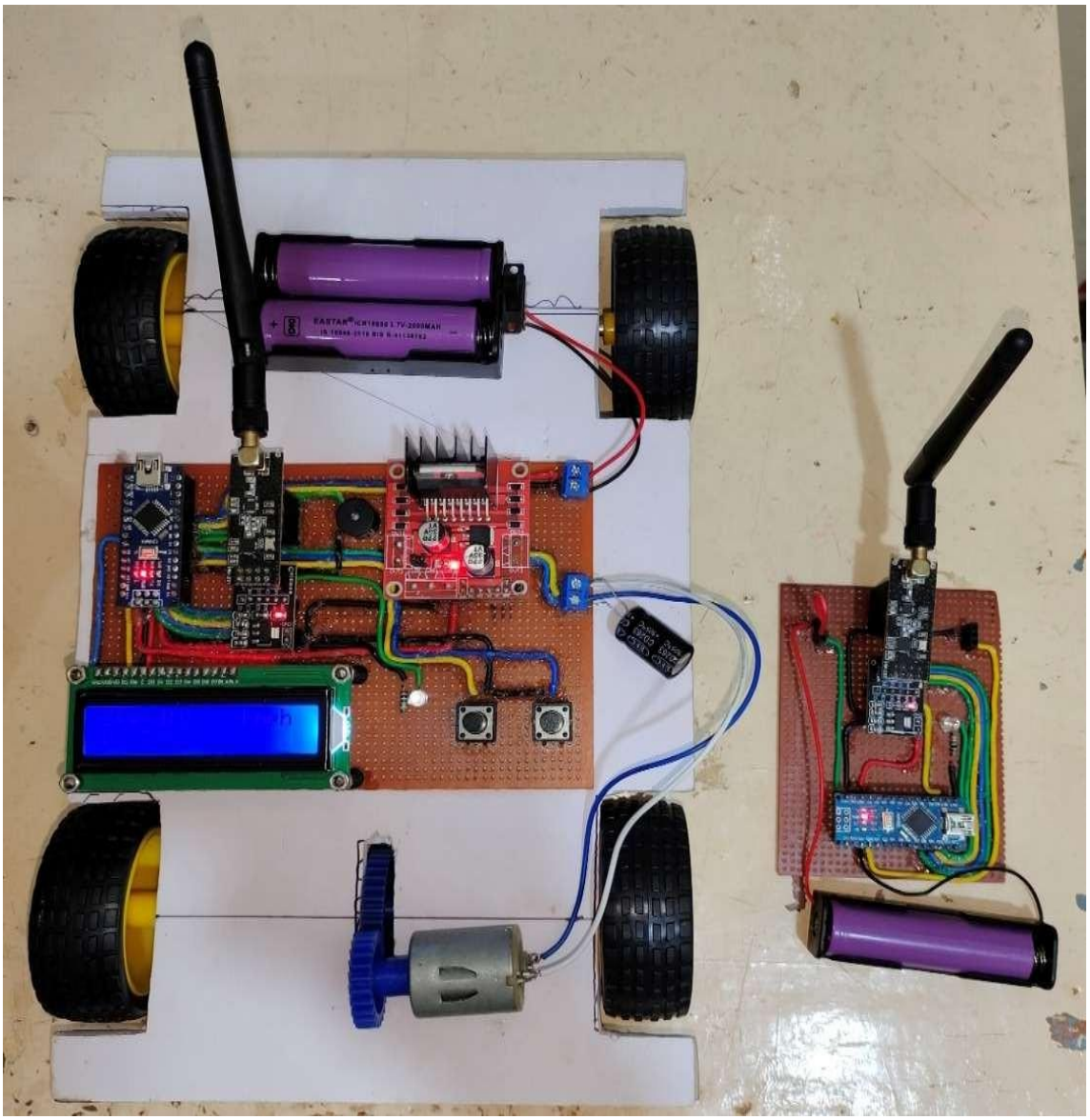

Fig : Project model

EXPLANATION:

The output of the **Automated Zone-Based Speed Limiter** system clearly demonstrates the successful communication between the transmitter and receiver modules and the effective automatic control of vehicle speed. When the transmitter switch is turned ON, it sends an RF signal (111) to indicate the presence of a restricted zone such as a school or hospital area. The LED on the transmitter glows to confirm that the signal is being broadcast. On the receiver side, when the RF module detects this transmitted signal, the microcontroller immediately identifies that the vehicle has entered a restricted zone. The buzzer produces a short alert sound, and the LED on the receiver glows, notifying the driver of zone entry. Simultaneously, the LCD display updates to show "School Zone" along with the current vehicle speed, confirming that the restricted zone has been detected.

Once the vehicle enters the restricted area, the system automatically begins to regulate the speed. If the current speed of the vehicle is higher than the predefined limit (for example, 30 km/h), the Arduino gradually reduces the motor speed using PWM control through the L298N driver until it reaches the safe limit. This reduction happens smoothly without sudden braking or jerks, ensuring both comfort and safety. The driver cannot increase the speed beyond the set limit while the vehicle remains within the zone, demonstrating the system's automatic intervention and reliability. When the RF signal is no longer detected (indicating the vehicle has exited the zone), the receiver resets to "No Zone" on the display, the LED turns off, and the driver regains manual speed control.

Overall, the observed output confirms that the system operates effectively under real-time conditions. It provides immediate detection of restricted zones, timely alerts to the driver, and controlled speed reduction — all of which contribute to enhanced road safety. The seamless communication between the transmitter and receiver and the accurate response of the motor speed to the control logic validate the efficiency and practicality of the proposed automated speed limiter for electric vehicles.

CHAPTER-4:

ADVANTAGES AND APPPICATIONS:

ADVANTAGES:

1. **Enhanced Road Safety**: Automatically controls vehicle speed in restricted zones, reducing accidents and ensuring pedestrian safety.
2. **Fully Automatic Operation**: Detects restricted zones and adjusts speed without ssdriver input, eliminating human error.
3. **Reliable and Fast Communication:** Uses nRF24L01 RF modules for quick and stable wireless response between transmitter and receiver.
4. **Energy Efficient and Cost-Effective:** Built with low-cost, low-power components like Arduino and RF modules, making it efficient and affordable.
5. **Smooth Speed Regulation**: Uses PWM for gradual speed control, providing comfort and reducing mechanical stress.
6. **Scalable and Upgradable:** Easily expandable and can integrate GPS, GSM, or IoT for smart monitoring and connectivity**.**
7. **Low Maintenance:** The system uses simple electronic components with minimal wear and tear, reducing maintenance needs**.**
8. **Environment Friendly**: Controlled driving and smooth speed regulation lead to less energy waste and lower emissions in hybrid or electric vehicles.
9. **Accurate Zone Detection:** Reliable RF communication ensures precise detection of speed zones, minimizing false triggers**.**
10. **Supports Smart City Integration:** Can be linked with intelligent transport systems and traffic management networks for enhanced urban safety.

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APPLICATIONS:

1. **School Zones:**

The system can be installed near schools to automatically limit the speed of vehicles passing through these areas, ensuring the safety of children during school hours.

2. **Hospital Areas:**

It helps maintain a calm and quiet environment near hospitals by automatically reducing vehicle speed and minimizing noise and vibration levels.

3. **Residential Areas:**

In densely populated residential zones, the system controls vehicle speed to reduce the risk of accidents, making streets safer for pedestrians and cyclists.

4. **Highways and Toll Booths:**

The technology can be used to manage vehicle speed near toll booths, entry/exit ramps, or accident- prone highway sections to prevent collisions and ensure smooth traffic flow.

5. **Industrial and Campus Zones:**

The system can be applied in large industrial areas, university campuses, or company premises where speed regulations are crucial for safety and discipline.

6. **Smart City Traffic Management:**

When integrated with IoT or GPS systems, this speed limiter can be part of a **smart transportation network**, allowing centralized monitoring and automatic speed regulation based on digital zones.

7. **Government and Public Transport Vehicles:**

It can be implemented in public buses, school vans, and government vehicles to ensure compliance with safety norms and reduce human error in speed control.

8. **Parking and Restricted Access Areas:**

The system can control speed within parking lots, airport runways, or gated communities where low- speed operation is necessary for safety and control.

CHAPTER 5:

CONCLUSION:

The **Automated Zone-Based Speed Limiter** system provides an innovative and practical approach to addressing one of the major causes of road accidents — over-speeding in restricted zones. By utilizissng **RF communication technology** and **Arduino-based control**, the system automatically detects predefined zones such as schools, hospitals, and residential areas and limits the vehicle's speed according to the set range. This automation ensures that vehicles comply with speed regulations without depending on the driver's attention or judgment, thereby enhancing safety and reducing human error.

The implementation of **NRF24L01 transceiver modules** enables reliable wireless communication between the transmitter (fixed at the zone entry) and the receiver (installed in the vehicle). The system successfully transmits zone signals and immediately triggers automatic speed control when the vehicle enters a restricted area. The use of **PWM (Pulse Width Modulation)** for speed regulation ensures smooth and gradual speed reduction, providing a comfortable driving experience without abrupt braking or mechanical strain on the vehicle's components. The **LCD display**, **LED indicator**, and **buzzer** provide clear visual and audio feedback to the driver, confirming proper system operation and zone detection.

From the experimental results, it was observed that the system responds instantly to zone changes and maintains stable communication within a range of **1 to 1.5 km**. The hardware implementation is simple, affordable, and consumes minimal power, making it suitable for **electric vehicles** and other low-energy systems. Compared to traditional methods such as speed breakers, signboards, and manual enforcement, this system offers a smarter, automated, and preventive solution rather than a corrective one.

In addition to improving road safety, the project demonstrates how embedded systems and wireless communication can be integrated to develop intelligent transportation technologies. It not only reduces the driver's workload but also contributes to the development of **smart mobility infrastructure**, which is an essential aspect of modern smart cities. The system's adaptability allows it to be customized for different types of zones with varying speed limits, ensuring versatile use in both urban and rural environments.

In the future, this project can be further improved by integrating **GPS, IoT, and cloud-based control systems** to dynamically detect and update speed zones in real time.

CHAPTER 8:

REFRENCES :

- Anam irdous, Indu, Vandana Niranjan, "*Zone Based Speed ontrol using R* " presented at 2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA) Galgotias University, Greater Noida, UP, India, October 2020. **https://ieeexplore.ieee.org/document/9250920** Dakshata.R. Dhote, Pragati.R. Lakde, Anjali.S. Pachare, Mukesh.A. Meharkure,Mahendra Dongare, "Design and Implementation of

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- Karthi eyan, Arshad Parvez, Kishore Kumar, Lavanya, Mathupra as, "R ID-Based Automatic Vehicle Speed ontrol" presented at International Journal of Science, JETIR, Volume 10, Issue 1, January 2023. **https://www.jetir.org/papers/JETIR2301580.pdf**

The Arduino code for the **Automatic Speed Control System using NRF24L01** was developed with reference to various open-source resources, including the Arduino official documentation https://www.arduino.cc/reference/en/, the RF24 wireless communication library by TMRh20 https://tmrh20.github.io/RF24/, and the LiquidCrystal I²C display library.

https://github.com/johnrickman/LiquidCrystal_I2C. This code enables a vehicle model to automatically adjust its motor speed based on wireless signals received from the NRF24L01 transmitter module, particularly in school zones. It integrates multiple components such as an LCD for displaying speed and zone status, a buzzer and LED for alerts, and motor control through PWM signals. When the receiver detects a specific signal (e.g., entering a school zone), the system limits the vehicle's speed automatically for safety and restores normal control after leaving the zone, demonstrating a practical application of embedded systems and wireless communication for intelligent traffic management.